

MEDICAL MONITORING AND ADVANCED SENSOR DATA PROCESSING – TASK 03: REAL-WORLD DATA ACQUISITION AND HANDLING

1 General Information

The practical experiments should be done in groups of up to three people. Each group must submit one solution via Digicampus. Of course, you can discuss the tasks between groups, but forwarding/copying other students' source code is forbidden.

Prepare the task until the deadline provided below/in Digicampus.

On Digicampus, there is a template for you to prepare your code for submission. Please complete your code within this template. The **README.md** file provides information on how to set up your Python environment for running your code. A Jupyter notebook example is also provided. You can adapt the content of this Jupyter notebook for preparing your solution.

Before uploading your solution, please make sure that the following requirements are met:

- Prepare your result documentation as Jupyter Notebook and a report
- Print a PDF version of your Jupyter notebook AND add it to the .zip archive you upload on Digicampus.
- All source code must be available and executable for us (this implies that you specify the required packages/libraries. In addition, you have to provide necessary data (ideally data should be in a subfolder, and you use relative paths) and/or give advice if paths have to be adjusted; in order to do so, you should provide a reasonable README file)
- **GREEN** sections in the chapter are tasks. Please include a numbering according to the questions' number Q<xy>
- List your sources (if applicable)

Submission deadline is **Thursday, 17.07.2025, 08:00**; submission is done via **Digicampus**.

Please ask in case of any doubts regarding the contents or the formal procedure.

2 Elements to be delivered

1. Jupyter Notebook: the students must deliver the code (as .ipynb file) they developed to visualize, analyse and plot the data according to the template provided.
A pdf version of the Jupyter notebook with the results of the notebook's run should be provided as a backup in case the notebook fails to be run again.

NOTE:

A requirements.txt file is provided on digicampus. Students HAVE TO USE ONLY the libraries that are provided within this file.

2. The students' recordings that are necessary to execute their Jupyter notebooks.
3. Report: the students must deliver a report (as .pdf file) using the ReportTemplate.docx found in Digicampus. The report must content the following:
Title Page
Title: Clear and specific
Course name / lab name
Students names
Date
Answer to the questions (with the appropriate question numbering)
4. When a question asks to discuss a topic, the word limit is an upper limit.

3 Remarks on the Hardware and software

A documentation to perform the recordings based on the smartwatch and Biopac platform is provided [1]. Executables to perform recordings via the ADI Study watch are provided in a Flashdrive [2]. Those executables should be copied to the computer which will record the data.

3.1 Biopac

One can export the acquired data in a format that is suitable for postprocessing.

3.2 ADI Study smartwatch

The communication between the smartwatch and the computer is established via a Dongle connected to a plugged USB port. One needs to check the port address (COM) using the device manager.

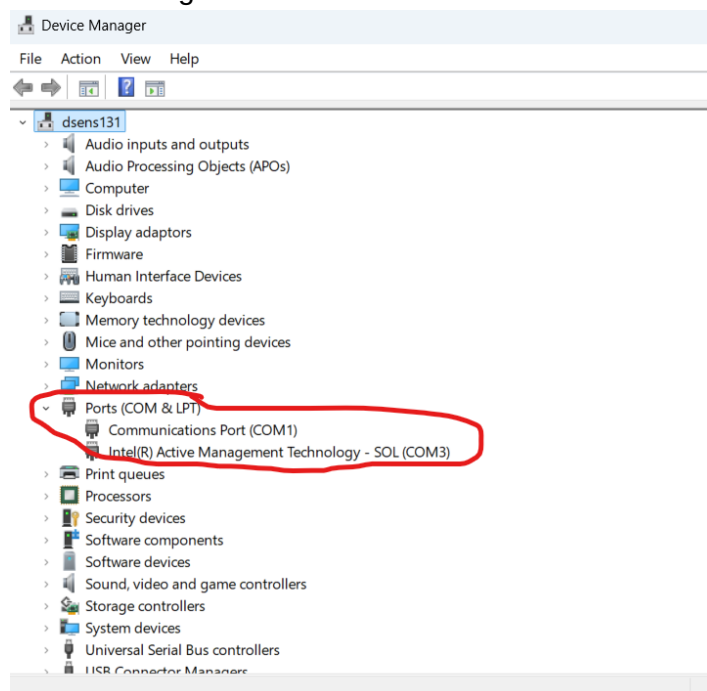


Figure 1 Device manager (on Windows).

The recordings are stored on the computer in .csv files.

The smartwatch can provide an electrocardiographic signal (ECG), a photoplethysmographic signal (PPG), and accelerometer signals (along the 3 directions).

4 Remarks on reports and data collection

Following some remarks on the report and data collection:

- Reports for data gathering should be detailed and complete. Students will use the data they collect to work on it by themselves.
- The report must assure the reproducibility of the experiment and data gathering, bearing in mind the amount and quality of data that is intended to be collected.
- If you cite from the literature, use the APA citation style
- For plots in the time domain, signals must be plotted with x-axis in time, in seconds.
- All plots must have axes labelled in a meaningful way.
- All plots or images must have caption written in a meaningful way.
- In the caption, do not forget to refer to the question number **Q...** the plot is built for.
- If a question asks you to make a plot, include the plots with appropriate reference to the question in the caption, and write a reference to the plot in the report.

For example :

Qxx. Area under the curve obtained on the test set by several classifiers is displayed on Figure 2.

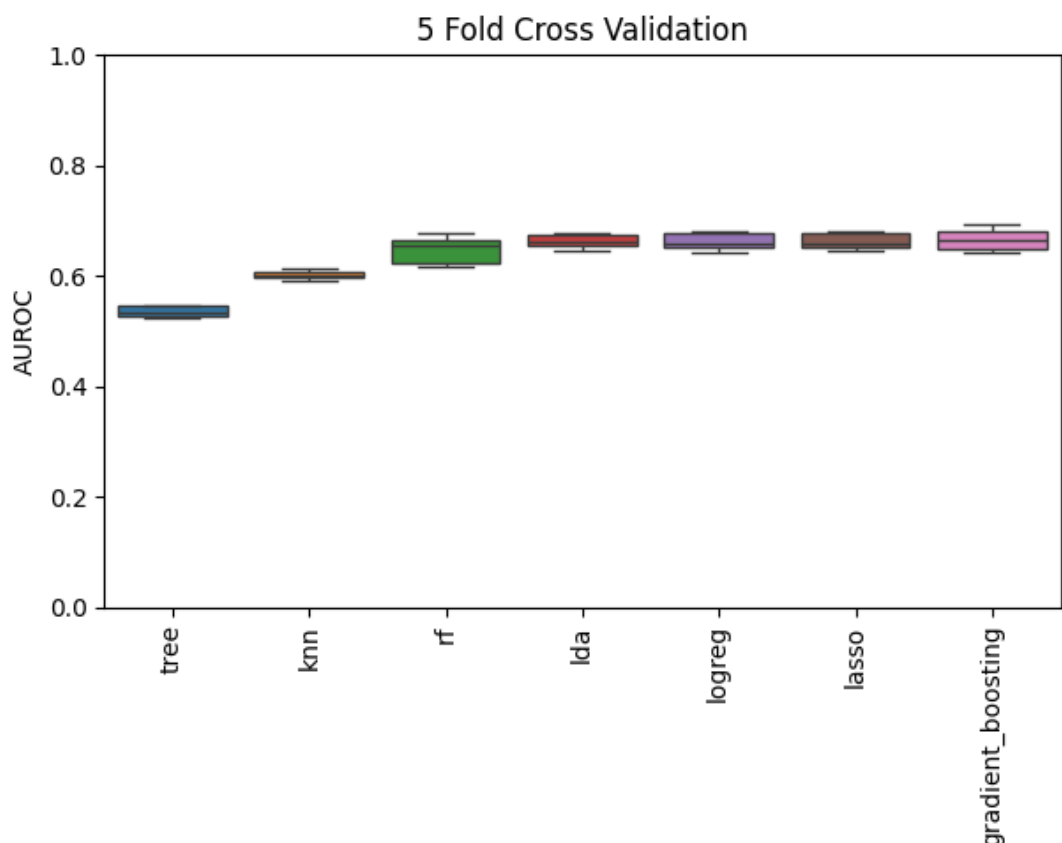


Figure 2 Area under the curve of several classifiers on the test set (**Qxx**).

- **ATTENTION:** Have a look at the general document on our expectation and common mistakes, which is available in Digicampus

5 Tasks

5.1 Notes

For the following tasks, you will need the measurements and scripts from Exercise 3. If anything was incorrect, please review the comments and correct any mistakes. If you have any questions about the comments or the previous task, don't hesitate to ask. Of course, you are also welcome to ask questions regarding this new exercise.

Unless a specific dataset is mentioned, complete the questions using both the "Stairs" and "Workout" data from both the Biopac and the smartwatch. You will not need your resting data for this exercise.

5.2 Questions

Q1 (1 Point)

- Synchronize your signals. Detect beats in all PPG and ECG signals and calculate the instantaneous heart rate (HR) (i.e. HR per beat) in beats per minute. Normalize the HR signals. Use the median smoother for all HR signals.
- For the filters you'll need ONE signal that represents the motion of the sensor (noise). Think about a way to combine the three acceleration channels into a single signal. Describe how you combined the channels by equations and a short description. Then, normalize the channel combination and each channel separately. Use your median smoother in order to have your motion signals corresponding to your HR signals (i.e. each signal one value starting at second 5).

Note: a) is what you did in your last Exercise (except for normalizing. You can extend your method from Exercise 3). You can and should use your code from last time. You can earn 0.5 points with this subtask. b) gives you 0.5 points.

Q2 (1.5 Points)

- Implement a Least Mean Squares (LMS) adaptive filter on your own solely using NumPy. Consider this task as the application of interference cancellation. What is your "Reference Signal"? What is your "System Input"? Choose your learning rate (typically below 0.1) and the number of filter coefficients. The equations from the lecture will help implement the filter.
- Filter all smoothed HR signals, denormalize (reverse your normalization) signals in order that the HR courses and MAE are back to interpretable values, plot HR courses in two plots, one for each measurement and calculate the MAE and Pearson correlation coefficient for all HR and for each workout/stair phase. Collect the results in a table. What differences do you notice? Discuss them.

Note: 1 Point is for a) the implementation – put your LMS method (it makes sense to write a method as you are filtering all signals with it) in your report and 0.5 is for b) the plots and results. Make sure that the plots and results are complete.

Q3 (1.5 Points)

- Implement a Recursive Least Squares (RLS) adaptive filter on your own solely using NumPy. Consider this task as the application of interference cancellation. The equations from the lecture will help implement the filter.
- What happens if you choose low values for the forgetting factor λ and the scaling factor δ ? What happens if you choose high values for forgetting factor λ and scaling factor δ ? Implement the filter and vary each parameter individually. Document your tests (you don't have to test the parameters using all data sets. Choose one) and discuss the results.
- Filter all smoothed HR signals, denormalize (reverse your normalization) signals in order that the HR courses and MAE are back to interpretable values, plot HR courses in two plots, one for each measurement and calculate the MAE and Pearson correlation coefficient for all HR and for each workout/stair phase. Extend the table with these results. What differences do you notice? Discuss them.

Note: 0.5 Point is for a) the implementation – put your RLS method (it makes sense to write a method as you are filtering all signals with it) in your report, 0.5 for b) the tests and questions and 0.5 is for c) the plots and results. Make sure that the plots and results are complete.

Q4 (2 Points)

- Implement a one-dimensional Kalman filter using only NumPy. The HR is the measurement you want to filter. You may assume that the initial uncertainty of system P is 1 at time 0, and that the process noise Q is very low (e.g., 0.001). Your model assumes that the HR does not change over time. Use the combined motion signal as the measurement noise (uncertainty of measurement) R.
- Filter all smoothed HR signals, denormalize (reverse your normalization) signals in order that the HR courses and MAE are back to interpretable values, plot HR courses in two plots, one for each measurement and calculate the MAE and Pearson correlation coefficient for all HR and for each workout/stair phase. Extend the table with these results. What differences do you notice? Discuss them.
- Choose one signal type (Biopac or Watch) and one measurement (Workout or Stairs) where you believe the Kalman filter performs well. Remember that you previously combined three motion signals into one. Now, take each of the original (normalized) motion signals individually and use them to filter the HR signal using the Kalman filter. Next, consider a scenario where no motion signals are available. In this case, generate a random noise signal and treat it as a substitute for the motion signal. Use it as the measurement noise and filter the HR signal. Plot the HR signals filtered with each individual motion signal and with the random noise, along with the corresponding ECG signal. Then, calculate the MAE and Pearson correlation coefficient for all HR signals, and separately for each phase. Summarize the results in a new table (it does not need to match the previous one). What differences do you notice? Discuss your findings. Can random noise effectively replace actual motion measurements?

Note: 0.5 Point is for a) the implementation – put your Kalman method (it makes sense to write a method as you are filtering a couple of signals with it) in your report, 0.5 is for b) the plots and results from a) and 1 point is for c). In c) you get 0.5 points for the part using single motion signals and 0.5 for the part using noise. Make sure that the plots and results are complete.

6 Material (and recommended readings)

- [1] Lecture 06 Adaptive Filter
- [2] Lecture 07 Kalman Filter